2021 TERM 2 PLANNER FOR YEAR 11 METHODS - SUBJECT TO CHANGE -
METHODS UNITS 1 & 2

Week	Lesson 1 - (MON P1 / MON P1)	Lesson 2 - (MON P2 / MON P2)	Lesson 3 - (WED P5 / WED P6)	Lesson 4 - (FRI P6 / FRI P5)
1 19 Apr	# Polynomials * 7C Factorisation of polynomials	# Polynomials * 7C Factorisation of polynomials	Trade-Off ?	# Polynomials * 7D Solving cubic equations
2 26 Apr	ANZAC Day	ANZAC Day	# Polynomials * 7F Graphs of factorised cubic functions Part 1	# Polynomials * 7E Cubic functions of the form f(x)=a(x-h)^3+k * 7F Graphs of factorised cubic functions
3 03 May	# Polynomials * 7G Finding rules for cubic functions	# Polynomials * 7I Applications of polynomial functions	# Probability * 9A Sample spaces & probability * 9B Estimating probabilities	# Probability * 9C Multi-stage experiements * 9D Combining events
4 10 May	# Probability * 9E Probability tables * 9F Conditional probability * 9G Independent events	# Probability * Ex 9E - 9G	# Counting Methods * 10A Addition and multiplication principles	# Counting Methods * 10B Arrangements
5 17 May	# Counting Methods * 10C Selections * 10D Applications to probability * 10E Pascal's triangle & the binomial theorem	# Counting Methods * Ex 10C - 10E	Methods Unit 1 Revision (Polynomials)	Methods Unit 1 Revision (Linear & Quadratics) Methods Unit 1 Revision (Transformations of functions)
6 24 May	S1 Exams	S1 Exams	S1 Exams	S1 Exams
7 31 May	S1 Exams	S1 Exams	S1 Exams	S1 Exams
8 7 Jun	WA Day	WA Day	Exam Review (CF)	Exam Review (CA)
9 14 Jun	# Differential Calculus * 17A Average rate of change	# Differential Calculus * 17B Instantaneous rate of change	# Differential Calculus * Ex 17A & 17B	# Differential Calculus * Sadler 5B **CAP: INV 2 MON**
10 21 Jun	# Differential Calculus * 17C The derivative	# Differential Calculus * 17D Rules for differentiation	# Differential Calculus * Sadler 5C & 5D	Trade-Off ?
11 28 Jun	# Differential Calculus * 17E Differentiating x^n (n is Z-)	# Differential Calculus * 17F Graphs of the derivative function	# Differential Calculus * 17G Antidifferentiation of polynomial	# Differential Calculus * Sadler Misc 5

Warm-Up: 3 minutes CF

- Divide $P(x) = x^3 + 4x^2 + x - 6$ by $x - 1$

- Divide $P(x) = x^3 + 4x^2 + x - 6$ by $x - 1$

$$
\begin{array}{r|l}
 & x^2 + 5x + 6 \\
\hline
x - 1 & x^3 + 4x^2 + x - 6 \\
 & -)\ x^3 - x^2 \\
\hline
 & 5x^2 + x \\
 & -)\ 5x^2 - 5x \\
\hline
 & 6x - 6 \\
 & -)\ 6x - 6 \\
\hline
 & 0
\end{array}
$$

$\therefore P(x) = (x-1)(x^2+5x+6)$

- Divide $P(x) = x^3 + 4x^2 + x - 6$ by $x - 1$

$$
\begin{array}{r@{}l}
 & x^2 + 5x + 6 \\
\cline{2-2}
x - 1 \,\big) & x^3 + 4x^2 + x - 6 \\
-) & x^3 - x^2 \\
\cline{2-2}
 & 5x^2 + x \\
-) & 5x^2 - 5x \\
\cline{2-2}
 & 6x - 6 \\
-) & 6x - 6 \\
\cline{2-2}
 & 0
\end{array}
$$

$$\therefore P(x) = (x-1)(x^2+5x+6)$$

$$= (x-1)(x+3)(x+2)$$

$$
\begin{array}{c|c}
x^2 & 6 \\
\hline
x & 3 \\
x & 2
\end{array}
$$

$$3x + 2x = 5x$$

YEAR 11 METHODS: POLYNOMIALS

Term 2 Week 1

7C Factorisation of polynomials

Part 1: Remainder theorem & Factor theorem

Learning Objectives/Syllabus Covered:

- 1.1.19 Factorise cubic polynomials in cases where a linear factor is easily obtained

7C Factorisation of polynomials

- Remainder theorem
- Factor theorem
- Rational-root theorem

$$a^2 + 2ab + b^2 = (a+b)^2$$

$$a^2 - 2ab + b^2 = (a-b)^2$$

$$a^2 - b^2 = (a+b)(a-b)$$

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Remainder theorem

Let $P(x) = x^3 + 3x^2 + 2x + 1$.

Divide $P(x)$ by $x - 2$:

▶ Remainder theorem

Let $P(x) = x^3 + 3x^2 + 2x + 1$.

Divide $P(x)$ by $x - 2$:

$$
\begin{array}{r}
x^2 + 5x + 12 \\
x - 2 \,\big)\, x^3 + 3x^2 + 2x + 1 \\
-)\ x^3 - 2x^2 \qquad\qquad \\
\hline
5x^2 + 2x \qquad \\
-)\ 5x^2 - 10x \qquad \\
\hline
12x + 1 \\
-)\ 12x - 24 \\
\hline
25
\end{array}
\rightarrow \text{Remainder}
$$

$\therefore P(x) = (x - 2)(x^2 + 5x + 12) + 25$

▶ Remainder theorem

Let $P(x) = x^3 + 3x^2 + 2x + 1$.

Divide $P(x)$ by $x - 2$:

$$
\begin{array}{r|l}
 & x^2 + 5x + 12 \\
x - 2 & x^3 + 3x^2 + 2x + 1 \\
 & -)\ x^3 - 2x^2 \\
 & \quad 5x^2 + 2x \\
 & -)\ 5x^2 - 10x \\
 & \quad\quad 12x + 1 \\
 & -)\ 12x - 24 \\
 & \quad\quad\quad 25
\end{array}
$$

$25 \to$ Remainder

$\therefore P(x) = (x - 2)(x^2 + 5x + 12) + 25$

Hence $P(2) = 0 + 25$
$= 25$

when $P(x)$ is divided by $x - \alpha$, the remainder is equal to $P(\alpha)$.

Remainder theorem

When $P(x)$ is divided by $\beta x + \alpha$, the remainder is $P\left(-\frac{\alpha}{\beta}\right)$.

Make $\beta x + \alpha = 0 \Rightarrow x = -\frac{\alpha}{\beta}$

CF

Example 9

Use the remainder theorem to find the value of the remainder when:

a $P(x) = x^3 - 3x^2 + 2x + 6$ is divided by $x - 2$

b $P(x) = x^3 - 2x + 4$ is divided by $2x + 1$.

CF

Example 10

When $P(x) = x^3 + 2x + a$ is divided by $x - 2$, the remainder is 4. Find the value of a.

Example 9

Use the remainder theorem to find the value of the remainder when:

a $P(x) = x^3 - 3x^2 + 2x + 6$ is divided by $x - 2$

b $P(x) = x^3 - 2x + 4$ is divided by $2x + 1$.

a) $P(2) = 2^3 - 3 \times 2^2 + 2 \times 2 + 6 = 6$ $\therefore$ remainder is 6

b) $2x + 1 = 0 \Rightarrow x = -\frac{1}{2}$

$\therefore P(-\frac{1}{2}) = (-\frac{1}{2})^3 - 2 \times (-\frac{1}{2}) + 4 = 4\frac{7}{8}$ $\therefore$ remainder is $4\frac{7}{8}$

Example 10

When $P(x) = x^3 + 2x + a$ is divided by $x - 2$, the remainder is 4. Find the value of a.

$P(2) = 2^3 + 2 \times 2 + a = 4$

$\therefore a = -8$

Factor theorem

For a polynomial $P(x)$:

- If $P(\alpha) = 0$, then $x - \alpha$ is a factor of $P(x)$.
- Conversely, if $x - \alpha$ is a factor of $P(x)$, then $P(\alpha) = 0$.

More generally:

- If $\beta x + \alpha$ is a factor of $P(x)$, then $P\left(-\frac{\alpha}{\beta}\right) = 0$.
- Conversely, if $P\left(-\frac{\alpha}{\beta}\right) = 0$, then $\beta x + \alpha$ is a factor of $P(x)$.

Factors of Polynomial $P(x)$

Consider the polynomial $P(x) = 3x^3 - 24x^2 + 57x - 36$.

Is $(x-3)$ a factor of $P(x)$?

Factors of Polynomial $P(x)$

Consider the polynomial $P(x) = 3x^3 - 24x^2 + 57x - 36$.

Is $(x-3)$ a factor of $P(x)$?

The Factor Theorem states that if $P(a) = 0$, then $x - a$ is a factor of $P(x)$.

$$
\begin{aligned}
P(3) &= 3 \times 3^3 - 24 \times 3^2 + 57 \times 3 - 36 \\
&= 3 \times 27 - 24 \times 9 + 171 - 36 \\
&= 81 - 216 + 171 - 36 \\
&= 252 - 252 \\
&= 0
\end{aligned}
$$

Hence $(x-3)$ is a factor of $P(x)$.

Using Long Division to Factorise Polynomial P(x)

Consider the polynomial $P(x) = 3x^3 - 24x^2 + 57x - 36$.

First : Use long division to find the quotient when $P(x)$ is divided by $x - 3$.

Using Long Division to Factorise Polynomial $P(x)$

Consider the polynomial $P(x) = 3x^3 - 24x^2 + 57x - 36$.

First : Use long division to find the quotient when $P(x)$ is divided by $x - 3$.

$$
\begin{array}{rrrrr}
 & 3x^2 & + -15x & + 12 & \\
\cline{2-5}
(x-3) \big) & 3x^3 & -24x^2 & +57x & -36 \\
-) & 3x^3 & -9x^2 & & \\
\cline{3-4}
 & & -15x^2 & +57x & \\
-) & & -15x^2 & +45x & \\
\cline{4-5}
 & & & 12x & -36 \\
 & & & 12x & -36 \\
\cline{4-5}
 & & & & 0
\end{array}
$$

Using Long Division to Factorise Polynomial $P(x)$

Consider the polynomial $P(x) = 3x^3 - 24x^2 + 57x - 36$.

First : Use long division to find the quotient when $P(x)$ is divided by $x - 3$: $(x-3)(3x^2-15x+12)$

Then : Factorise the quadratic $3x^2 - 15x + 12$

$$\begin{array}{c|c} 3x^2 & 12 \\ \hline 3x & -3 \\ x & -4 \end{array}$$

$$(3x-3)(x-4).$$

$$-12x - 3x = -15x \checkmark$$

Using Long Division to Factorise Polynomial $P(x)$

Consider the polynomial $P(x) = 3x^3 - 24x^2 + 57x - 36$.

First : Use long division to find the quotient when $P(x)$ is divided by $x - 3$: $(x-3)(3x^2-15x+12)$

Then : Factorise the quadratic $3x^2 - 15x + 12$:

$$\begin{array}{c|c} 3x^2 & 12 \\ \hline 3x & -3 \\ x & -4 \end{array}$$

$$(3x-3)(x-4).$$

$$-12x - 3x = -15x \checkmark$$

Finally: Write the polynomial in factorised form:

$$P(x) = (x-3)(\underline{3}x-\underline{3})(x-4) = 3(x-3)(x-1)(x-4)$$

- Remainder theorem When $P(x)$ is divided by $\beta x + \alpha$, the remainder is $P\left(-\frac{\alpha}{\beta}\right)$.
- Factor theorem
  - If $\beta x + \alpha$ is a factor of $P(x)$, then $P\left(-\frac{\alpha}{\beta}\right) = 0$.
  - Conversely, if $P\left(-\frac{\alpha}{\beta}\right) = 0$, then $\beta x + \alpha$ is a factor of $P(x)$.

Important * A cubic function can be factorised using the factor theorem to find the first linear factor and then using polynomial division or the method of equating coefficients to complete the process.

CF

Example 12

Factorise $x^3 - 2x^2 - 5x + 6$.

Important

* A cubic function can be factorised using the factor theorem to find the first linear factor and then using polynomial division or the method of equating coefficients to complete the process.

CF

Example 12

Factorise $x^3 - 2x^2 - 5x + 6$.

Let $P(x) = x^3 - 2x^2 - 5x + 6$

Use Factor Theorem: $P(1) = 1 - 2 - 5 + 6 = 0$ ← Often try $x = 1, 2, 3, 4, 5 \ldots$

$-1, -2, -3, -4, -5 \ldots$

$\therefore (x-1)$ is a factor of $P(x)$

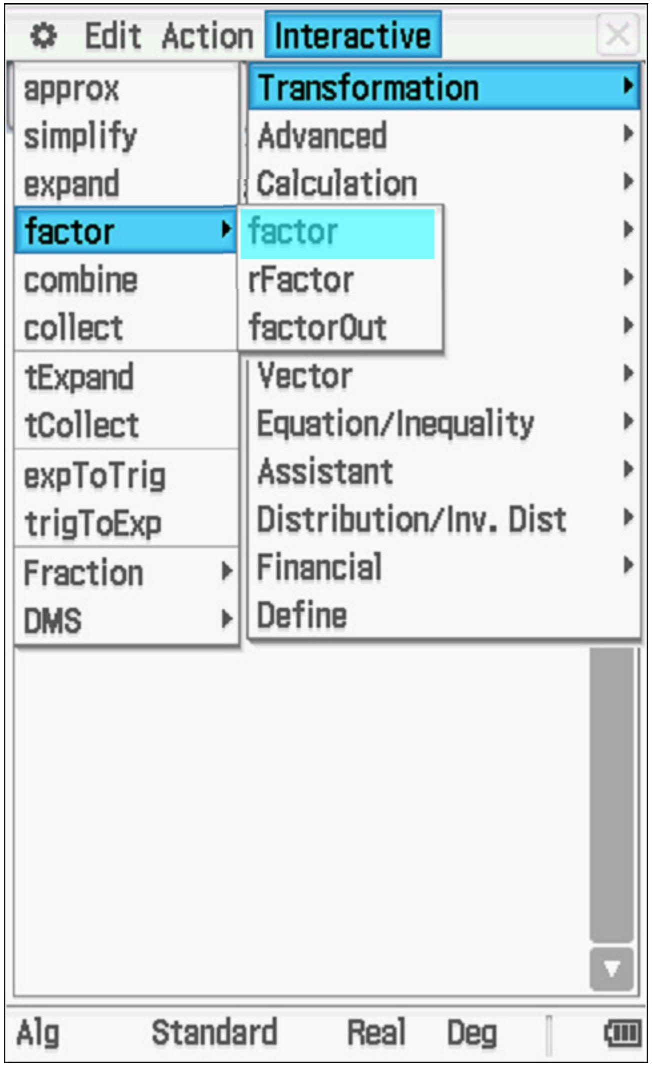
$$\therefore \quad \begin{array}{r|l} & \quad x^2 - x - 6 \\ \hline x-1 & x^3 - 2x^2 - 5x + 6 \\ -) & x^3 - x^2 \\ \hline & \quad -x^2 - 5x \\ -) & \quad -x^2 + x \\ \hline & \qquad -6x + 6 \\ -) & \qquad -6x + 6 \\ \hline & \qquad\qquad 0 \end{array}$$

$$\therefore P(x) = (x-1)(x^2 - x - 6)$$

$$= (x-1)(x-3)(x+2)$$

$$\begin{array}{c|c} x^2 & -6 \\ \hline x & -3 \\ x & 2 \end{array}$$

$-3x + 2x = -x$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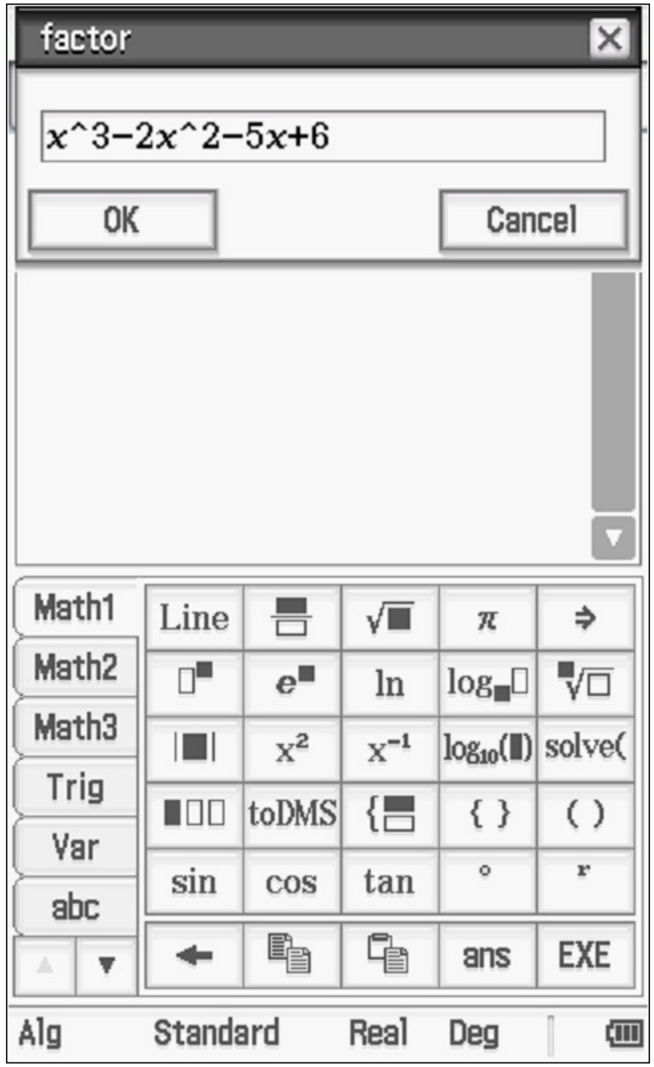
factor
x^3−2x^2−5x+6
OK
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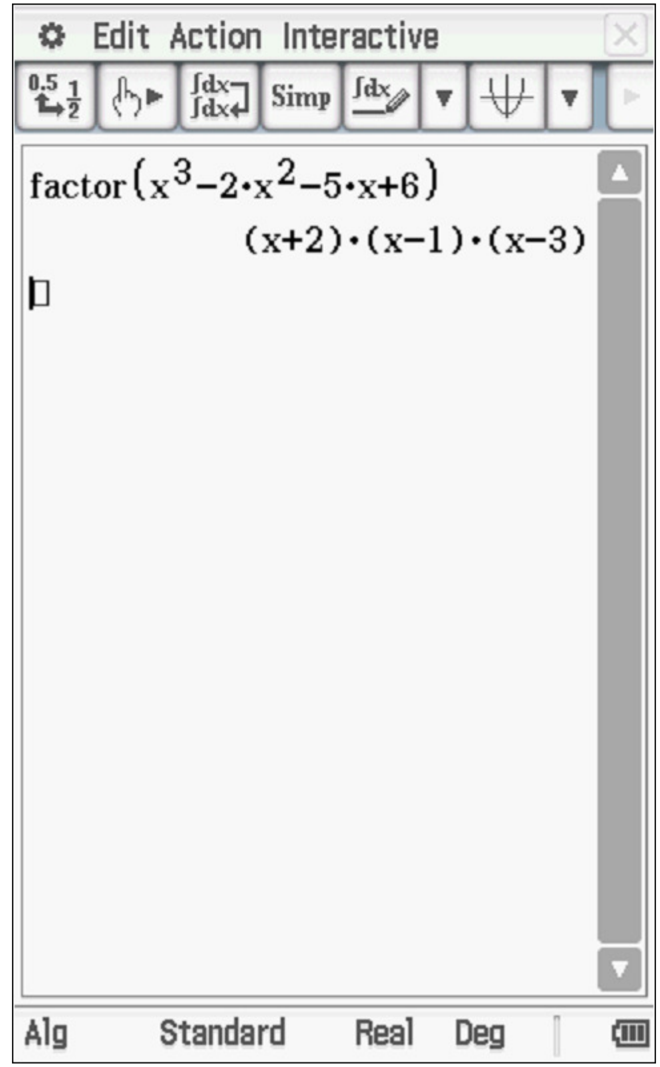
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Simp
factor(x^3−2·x^2−5·x+6)
(x+2)·(x−1)·(x−3)
Alg Standard Real Deg

Practice : $x+2$ is a factor of $P(x) = x^4 + 3x^3 - 7x^2 - 27x - 18$. Factorise $P(x)$.

Practice : $x+2$ is a factor of $P(x)=x^4+3x^3-7x^2-27x-18$. Factorise $P(x)$.

$$x^3 + x^2 + (-9x) + (-9)$$

$$(x+2)\,\overline{)\,x^4+3x^3-7x^2-27x-18}$$

$$-)\ \underline{x^4+2x^3}$$

$$x^3-7x^2$$

$$-)\ \underline{x^3+2x^2}$$

$$-9x^2-27x$$

$$-)\ \underline{-9x^2-18x}$$

$$-9x-18$$

$$-)\ \underline{-9x-18}$$

$$0$$

Practice:

$x+2$ is a factor of $P(x)=x^4+3x^3-7x^2-27x-18$

$= (x+2)(x^3+x^2-9x-9)$

$= (x+2)(x^3-9x+x^2-9)$

Practice:

$x + 2$ is a factor of $P(x) = x^4 + 3x^3 - 7x^2 - 27x - 18$

$$= (x+2)(x^3 + x^2 - 9x - 9)$$

$$= (x+2)(x^3 - 9x + x^2 - 9)$$

$$= (x+2)\left[x(x^2-9) + (x+3)(x-3)\right]$$

$$= (x+2)\left[x(x+3)(x-3) + (x+3)(x-3)\right]$$

$$P(x) = (x+2)(x+3)(x-3)(x+1)$$

Assigned Tasks

- **Cambridge Methods Maths:**
- **Exercise 7C** (Q1 – Q6)

Mathspace

DDFKY8	https://mathspace.co/join/

1. Equations and Polynomials > 1.09 Factorise cubic polynomials
Adaptive task

Duration
Mon 19 Apr 9:45am - Sun 25 Apr 2:00pm